Reemika Subrata Das

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PROFESSIONAL SUMMARY

Data Analytics and AI professional with an M.S. in Business Analytics (AI/ML) and 4+ years of BI and MIS analytics experience in mortgage and finance. Built and deployed a Text-to-SQL assistant over 111K+ records that cut investigation time ~90% and LLM spend 43%, plus forecasting models (2.4% MAPE at Adobe) and RAG applications. Skilled in SQL, Python, Tableau, Databricks, and LLM evaluation.

TECHNICAL SKILLS

Languages & Querying: Python (Pandas, NumPy, PySpark), SQL (MySQL, PostgreSQL; joins, CTEs, window functions), Advanced Excel

Data Analysis & Statistics: Exploratory Data Analysis, Time-Series Analysis (ADF/KPSS stationarity testing), Hypothesis Testing, Statistical Modeling (Statsmodels), Data Validation & Cleaning, KPI Reporting

BI & Visualization: Tableau, Looker Studio, Matplotlib, Excel Dashboards

Data Engineering: ETL/ELT Pipelines, Databricks, Delta Lake, Unity Catalog, Auto Loader, Medallion Architecture, Schema Validation

Machine Learning & Forecasting: Scikit-learn, TensorFlow, Keras, PySpark ML, Feature Engineering, Supervised Learning, Model Evaluation (MAPE, MAE, R^2, Accuracy, Precision, Recall)

Generative AI & LLM Applications: Retrieval-Augmented Generation (RAG), Text-to-SQL, Prompt Engineering, LangChain, FAISS, Hugging Face, LangSmith, LLM Evaluation (LLM-as-judge), LLM Observability & Cost Optimization

Cloud, APIs & Dev Tools: AWS EC2, FastAPI, REST APIs, Streamlit, Git/GitHub, GitHub Actions, SQLGlot, Claude Code, Codex

PROFESSIONAL EXPERIENCE

Santa Clara University

Research Assistant | Dec 2025 - Present

- Reduced provenance-record investigation by approximately 90% by building and deploying a Python-based AI assistant on AWS EC2 that translates natural-language questions into inspectable MySQL queries across 111K+ records
- Secured the Text-to-SQL workflow with SQLGlot validation, table allowlisting, SELECT-only enforcement, automated tests, and model fallback, enabling migration from Mistral to Qwen while evaluating DeepSeek as an alternative
- Built a Looker Studio observability layer with a 15-minute dashboard refresh, transforming OpenRouter logs into BI-ready datasets for tracking token usage, caching, latency, truncation, and cost per request
- Reduced observed LLM cost per request by 90% and total spend by 43% while handling 5.9× more requests at comparable latency, based on analysis of 1,850+ requests and 4.24M tokens

Adobe (with Santa Clara University)

Data Scientist Practicum | Jan 2025 - Jun 2025

- Applied statistical analysis and machine learning techniques to develop, evaluate, and benchmark ARIMA, feed-forward neural network, RNN, and LSTM models, achieving 2.4% MAPE for subscription-seat demand forecasting
- Built a reusable Python data-processing and ML pipeline across 6,400+ structured daily records, engineering lag, rolling-window, fiscal-calendar, holiday, growth, and Fourier features to model trend and seasonality
- Prepared and validated modeling datasets using Python, Pandas, NumPy, statistics, and probability, applying missing-value treatment and ADF/KPSS tests to assess stationarity and support reliable model development

XL Dynamics India Pvt Ltd

Process Executive MIS | Nov 2021 - Aug 2024

- Prepared 30+ MIS reports for senior management, finance, and marketing teams by extracting, cleaning, validating, and transforming structured mortgage data using Excel VLOOKUP, PivotTables, logical formulas, and filters
- Created impactful Excel dashboards and interpreted data for reporting across 7-8 cross-functional teams to deliver decision-ready analysis
- Performed ad hoc analysis to translate business requirements into analytical solutions and reduced knowledge gaps and escalation by creating a Standard Operating Procedure (SOP) to carry out routine operations

MP Financial Advisory Services LLP

Credit Risk Analyst | Jun 2021 - Oct 2021

- Built Excel financial models for 5 client credit evaluations, summarizing creditworthiness, risk exposure, and recommendation drivers

Trio Logistics Pvt Ltd

Account Assistant | Jun 2018 - Jun 2019

- Improved tax-filing readiness for 50+ clients by preparing reconciliation reports, monthly summaries, and accurate financial records

SELECTED PROJECTS

Uber vs Lyft Fare Analytics Platform | Databricks ELT, Tableau BI & Machine Learning

- Engineered a governed Databricks medallion ELT pipeline for 699,347 fare and weather records using PySpark, Auto Loader, Delta Lake, and Unity Catalog; validated and retained 92% of fare quotes and achieved 100% endpoint-weather matching
- Analyzed 425,321 comparable fare quotes across 72 directional routes through an interactive Tableau dashboard, identifying lower median price per mile for Lyft in Standard and XL tiers and for Uber in both Premium Black tiers
- Developed leakage-safe regression models in PySpark ML; selected Gradient-Boosted Trees over the tuned Random Forest for greater efficiency, improving MAE by 32% over Linear Regression and achieving a $1.19 test MAE and 0.9625 R^2

Coding Challenge Publisher | MySQL, PostgreSQL, Python, GitHub API & GitHub Actions

- Solved and documented 180+ SQL and Python challenges (DataLemur, HackerRank, LeetCode) covering joins, CTEs, subqueries, window functions, and ranking
- Built an automated tool that publishes standardized solutions to GitHub, cutting manual publishing effort by approximately 90%

HealthMate: RAG Fitness & Nutrition Chatbot | Python, LangChain, LangSmith, Hugging Face, FAISS, Groq

- Designed and developed a full-stack Retrieval-Augmented Generation (RAG) solution over eight curated knowledge documents using React, TypeScript, Python, FastAPI, LangChain, Hugging Face embeddings, FAISS vector search, semantic retrieval, and Groq LLM inference to produce grounded, cited responses
- Created an AI evaluation methodology with 36 benchmark test cases and LangSmith/LLM-as-judge workflows to measure correctness, relevance, groundedness, retrieval quality, and end-to-end RAG performance

EDUCATION

Santa Clara University - M.S. in Business Analytics | GPA: 3.71/4.00 | Sept 2024 - Dec 2025
ITM Business School - Post Graduate Diploma in Management | GPA: 3.02/4.00 | Jun 2019 - Jun 2021
University of Mumbai - Bachelor's in Management Studies | GPA: 3.70/4.00 | Jun 2015 - Apr 2018